Action Recognition of Persons in Aerial Videos (Okutama Dataset)

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MTP Phase-I Presentation

Topics to be covered

- Introduction & Motivation
- Problem Statement & Project Goal
- Dataset Overview & Visualization
- Baseline Approaches (YOLO, Swin, TimeSformer)
- Proposed Pipeline: YOLOv8 + Hybrid Tracking + TimeSformer
- Experimental Results
- Conclusion & Future Work

Introduction & Problem Statement

Introduction

- Aerial videos captured by UAVs are increasingly used in surveillance, traffic monitoring, disaster response, and large-scale human activity analysis.
- Unlike ground-level videos, UAV footage presents unique challenges:
  - Extremely small human appearance due to high altitude
  - Rapid drone motion, vibration, and camera shake
  - Frequent occlusions and complex backgrounds
  - Multiple people performing different actions simultaneously
- These factors make accurate person detection, tracking, and action recognition significantly more difficult.

Motivation

- Most action recognition models work well on ground videos where people are big and clear.
- But in aerial videos:
  - People appear very small
  - Drone motion creates a lot of background movement
  - Many people perform different actions at the same time
- Because of these challenges, we need a **person-specific** and **robust** system that can:
  - Detect each person clearly
  - Track them over time
  - Recognize their actions accurately

Problem Statement & Project Goal

- **Problem Statement:** How can we automatically detect individuals, maintain their identities over time, and understand their actions accurately in complex, real-world aerial videos captured by UAVs?
- **Project Goal:**
  - Develop an end-to-end system capable of:
    - Detecting people in drone-captured aerial frames,
    - Tracking their identities robustly over time,
    - Recognizing multiple actions for each person at the same time.

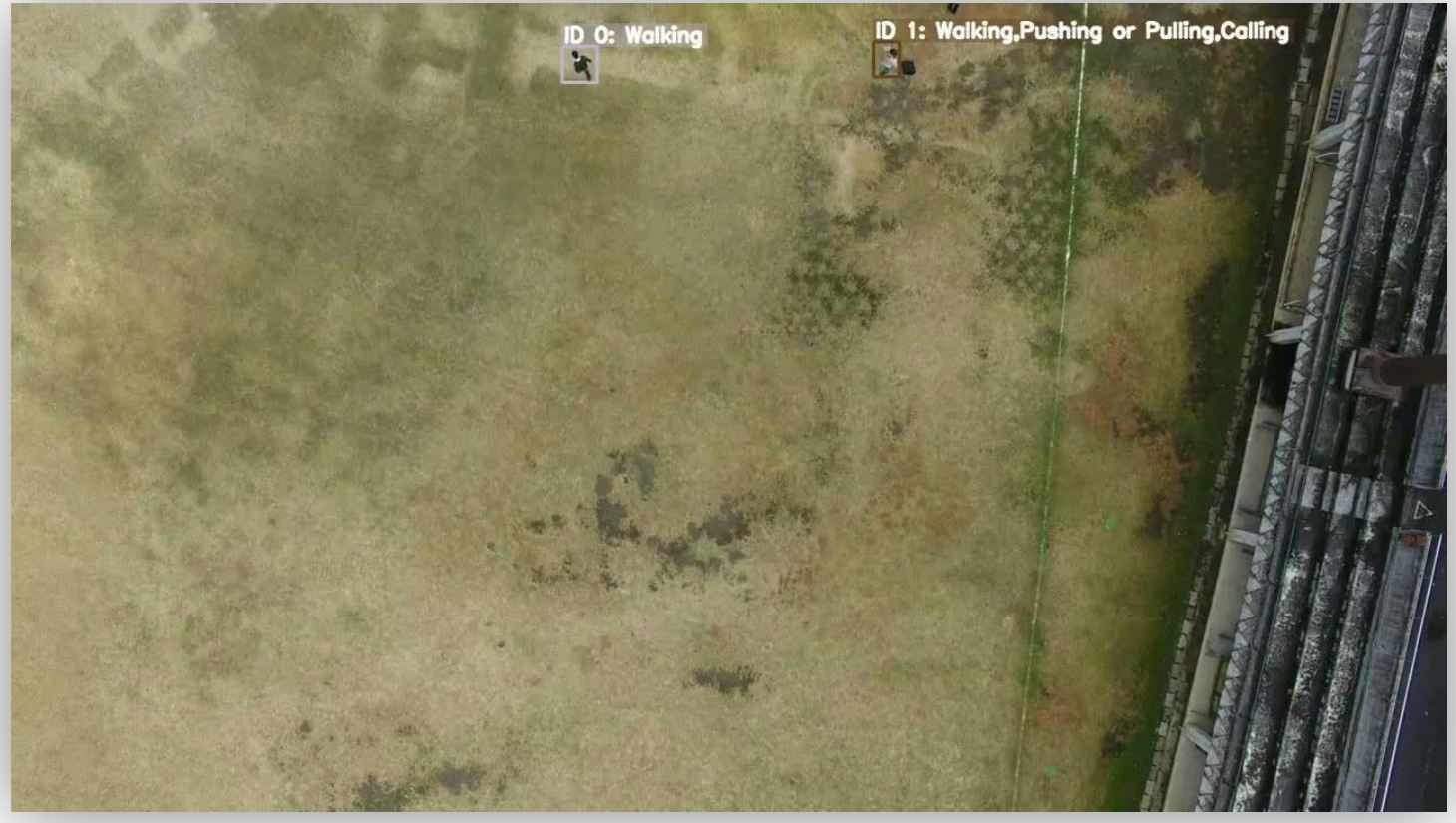
Dataset Overview & Visualization

Dataset: Okutama-Action

UAV videos captured using **DJI Phantom 4** drone.

- **43 minutes** of 4K UAV video footage
- **Resolution:** 3840×2160 (4K)
- **Frame rate:** 30 FPS
- **Drone altitude:** 5-10 meters
- **12 human actions:**
  - Calling, Carrying, Drinking
  - Handshaking, Hugging, Lying
  - Pushing/Pulling, Reading
  - Running, Sitting, Standing, Walking
- Videos converted to frames and annotations parsed

Real Dataset Visualization



Real Dataset Visualization

Overview of Work Presented

- **Studied multiple baseline approaches and their limitations:**
  - YOLOv8 person detection
  - YOLO + Swin Transformer
  - YOLO + TimeSformer (scene-level)
  - YOLO + TimeSformer (person-level)
- **Designed and implemented the final pipeline:**
  - YOLOv8 for person detection
  - Hybrid DeepSORT + Particle Filter for tracking
  - TimeSformer for person-specific action recognition

What is YOLO?

- YOLO (You Only Look Once) is a real-time object detection framework.
- Unlike two-stage detectors (Faster R-CNN), YOLO performs:
  - **Single forward pass** to detect all objects
  - **Direct regression** of bounding boxes + class probabilities
- Divides the image into grids and predicts:
  - Object presence
  - Bounding box coordinates
  - Class label
- Designed for high-speed, edge-friendly deployment (perfect for UAV applications).

Advantages & Limitations of YOLO

Advantages:

- **Fast and real-time** — single forward pass.
- **Good accuracy** for person detection in aerial videos.
- **Anchor-free (YOLOv8)** improves small person detection.
- Handles motion blur and scale variation reasonably well.

Limitations:

- Only detects objects — **no temporal information**.
- Cannot classify actions (walking, running, sitting, etc.).
- Performance drops when persons are extremely small.
- One-label-per-box only $\rightarrow$ **not multi-label** for actions.

What is Swin Transformer?

- Swin Transformer (Shifted Window Transformer) is a hierarchical vision transformer.
- Designed to handle:
  - Long-range dependencies
  - High-resolution images
  - Multi-scale visual features
- Uses **shifted window attention**:
  - Local attention for efficiency
  - Shifted windows for cross-window context
- Achieves strong performance in classification, detection, and segmentation.

Limitations of Swin for Aerial Action Recognition

- Works on the **full frame**, so actions of different people get mixed.
- **Humans look very small** in aerial videos $\rightarrow$ Swin loses important details.
- Drone motion adds background noise and confuses the model.
- Many actions (walking, running, standing) look similar from top view.
- Swin focuses mainly on spatial features and **cannot capture motion well**.
- Leads to:
  - Lower accuracy
  - Same action predicted for all persons

What is TimeSformer?

- TimeSformer is a transformer-based video action recognition model.
- Introduces **Divided Space–Time Attention**:
  - Spatial attention (per frame)
  - Temporal attention (across frames)
- Processes video clips as sequences of image patches.
- Captures long-range temporal motion better than CNNs or 3D Conv models.
- Strong performance on large-scale datasets (Kinetics, Something-Something).

Limitations of TimeSformer (Scene Level)

- Operates on the **full video frame** $\rightarrow$ mixes actions of all persons.
- **Drone motion** adds significant background noise.
- Very **small human size** makes temporal attention ineffective.
- Produces **one label per video clip** $\rightarrow$ not suitable for multi-person scenes.
- Confusion between visually similar aerial actions:
  - Walking vs Standing vs Running
- Result:
  - Low accuracy
  - No person-specific action recognition

TimeSformer for Person-Level Action Recognition

- Each person is cropped using YOLO detections + Hybrid tracking.
- Tracklets of **16 frames** (224×224) are fed into TimeSformer.
- Removes background noise and drone motion.
- Enables:
  - **Person-specific** action recognition
  - **Multi-label** prediction for each individual
  - Clear temporal patterns (walking, running, sitting, etc.)
- Significantly higher reliability compared to scene-level classification.

System Architecture

Input Aerial Video

↓

YOLOv8 Detection
Detect persons (bounding boxes)

↓

Hybrid Tracking
DeepSORT + Particle Filter

↓

Tracklet Buffer
16-frame person clips

↓

TimeSformer
Person-level actions

↓

Final Output
ID + Actions

Description

- **Input Aerial Video** — 4K UAV footage at 30 FPS.
- **YOLOv8 Detection** — Locates all persons in each frame.
- **Hybrid Tracking** — DeepSORT appearance + Particle Filter motion ensure ID stability.
- **Tracklet Buffer** — Stores short per-person clips (e.g., 16 frames).
- **TimeSformer** — Classifies person-specific actions using temporal attention.
- **Final Output** — Stable Track IDs + multi-label action predictions.

Results & Visualizations

YOLOv8 Results

Metric	Value
Precision	0.9199
Recall	0.7582
mAP@0.5	0.8324
mAP@0.5:0.95	0.4796

Key Points:

- **Precision 92%**: Very few false detections.
- **Recall 76%**: Most persons detected.
- **mAP@0.5 = 0.83**: Strong detection performance.
- **mAP@0.5:0.95 = 0.48**: Decent for small aerial humans.

Observations:

- Works reliably under motion blur and illumination variation.
- Runs at **25–30 FPS** on GPU $\rightarrow$ near real-time.

Yolo Visualization

TimeSformer Results

Metric	Value
Accuracy	0.2193
Macro F1-score	0.1964
Micro F1-score	0.4722

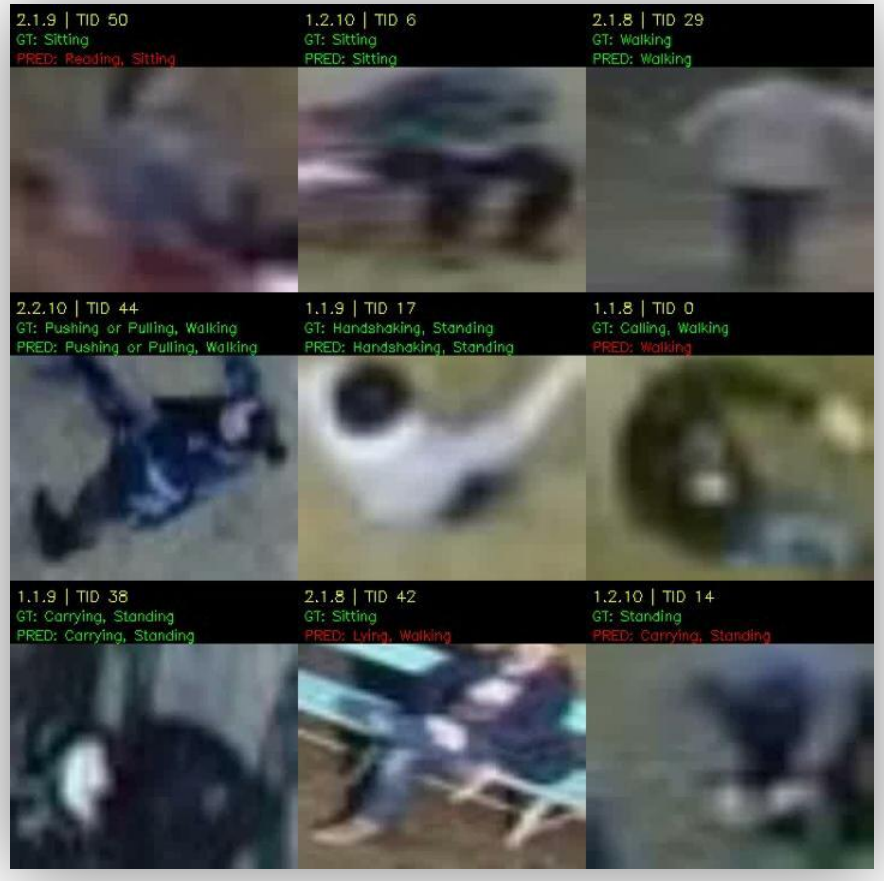
What the metrics mean:

- **Accuracy**: Overall percentage of correctly predicted action labels.
- **Macro F1-score**: Treats all classes equally (average F1 per action). Good for checking performance on rare actions.
- **Micro F1-score**: Aggregates all predictions across all classes. Better for multi-label tasks and imbalanced datasets.

Insights:

- **Good at:** Standing, Walking, Running — strong temporal motion cues.
- **Hard cases:** Calling, Reading — very small regions in aerial videos.

TimeSformer Person Level Visualization



TimeSformer Person Level Visualization

Final Pipeline Visualization

Final Pipeline Visualization

Why FPS is Low in the Current Pipeline

Major Bottlenecks:

- **YOLOv8** is computationally heavy for high-resolution aerial frames.
- **TimeSformer** requires processing 16-frame clips $\rightarrow$ high GPU load.
- Both modules are run frequently, reducing throughput.
- **Tracker (Hybrid DeepSORT + Particle Filter)** is fast:
  - Purely mathematical updates
  - Can run on every frame (**low computation cost**)
- Therefore:
  - Detection + Action Recognition slow down FPS
  - Tracking alone does not affect speed

Improving FPS with Smart Inference Strategy

- **Run YOLOv8 + TimeSformer only on the first frame** Heavy models run less often → big FPS boost.
- **Tracker runs on all remaining frames** Lightweight and keeps identities stable in real time.
- **Re-run full detection only when needed:**
  - Person going out of frame (80%)
  - Low tracker confidence
  - New person entering the scene
- **Extra Strategies to Maintain Accuracy:**
  - **Run YOLO more often when needed** (for example, when people move fast or tracker is not confident).
  - **Refresh detection after long occlusions** so the tracker does not lose the person.
  - **Re-run YOLO if actions suddenly change** to avoid wrong predictions.

How to Increase FPS for Real-Time Performance

```
[ACTION] Frame 280 | Track 0 -> ['Standing', 'Walking']
[ACTION] Frame 280 | Track 1 -> ['Standing', 'Walking']
[ACTION] Frame 280 | Track 2 -> ['Standing', 'Pushing or Pulling']
[ACTION] Frame 280 | Track 3 -> ['Standing', 'Walking']
[ACTION] Frame 280 | Track 4 -> ['Standing', 'Walking']
[ACTION] Frame 280 | Track 6 -> ['Standing', 'Walking']
[ACTION] Frame 288 | Track 0 -> ['Standing', 'Walking']
[ACTION] Frame 288 | Track 1 -> ['Standing', 'Walking']
[ACTION] Frame 288 | Track 3 -> ['Standing', 'Walking']
[ACTION] Frame 288 | Track 4 -> ['Standing', 'Walking']
[ACTION] Frame 288 | Track 6 -> ['Standing', 'Walking']
[ACTION] Frame 296 | Track 0 -> ['Standing', 'Walking']
[ACTION] Frame 296 | Track 1 -> ['Standing', 'Walking']
[ACTION] Frame 296 | Track 3 -> ['Standing', 'Walking']
[ACTION] Frame 296 | Track 4 -> ['Standing', 'Walking']
[ACTION] Frame 296 | Track 6 -> ['Standing', 'Walking']
[INFO] Processed 300 frames in 14.19s => 21.14 FPS
[INFO] Saved tracking + actions CSV → /home1/jtt_1/mtp/outputs/2.2.1/2.2.1_tracking.csv
```

Conclusion

- Developed a complete **per-person action recognition pipeline** for aerial videos.
- **YOLOv8**: Achieved high-precision person detection under UAV conditions.
- **Hybrid DeepSORT + Particle Filter**: Provided stable identity tracking with fewer ID switches.
- **TimeSformer**: Delivered improved person-level multi-label action recognition.
- Overall system handles drone motion, occlusions, and multi-person scenarios effectively.

Future Work

- **Develop a more robust hybrid tracking algorithm** capable of maintaining consistent identities even when a person leaves and re-enters the frame.
- **Improve the action recognition pipeline** to better handle fine-grained and small-scale aerial actions.
- **Integrate Swin Transformer with TimeSformer** to build a stronger spatial–temporal feature extraction framework for more reliable action prediction.
- **Increase FPS while retaining accuracy** by optimizing detection frequency, using lightweight models, and exploring keyframe-based action prediction strategies.

Thank you!